Enderton's Elements of Set Theory

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Chapter 1

Axioms and Operations

1.1 First Axioms

Extensionality axiom
If two sets have the exact same members, then they are equal:

$$\forall A \forall B [\forall x (x \in A \leftrightarrow x \in B) \rightarrow A = B]$$

(The nested implication is not an equivalence because the other direction is already true. Declaring it would be redundant.)

Empty set axiom
There is a set having no members:

$$\exists B \forall x (x \notin B)$$

Pairing axiom
For any sets u and v, there is a set having as members just u and v:

$$\forall u \forall v \exists B \forall x (x \in B \leftrightarrow x = u \text{ or } x = v)$$

Union axiom, Preliminary form
For any sets a and b, there is a set whose members are those sets belonging either to a or to b (or both):

$$\forall a \forall b \exists B \forall x (x \in B \leftrightarrow x \in a \text{ or } x \in b)$$

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Power set axiom
For any set a, there is a set whose members are exactly the subsets of a:

$$\forall a \exists B \forall x (x \in B \leftrightarrow x \subseteq a)$$

We can rewrite $x \subseteq a$ in terms of the definition of $\subseteq$:

$$\forall t (t \in x \to t \in a)$$

This list will be further expanded later to include

- Subset axioms
- Replacement axioms
- Infinity axiom
- Regularity axiom
- Choice axiom

The union axiom will be also be restated in a stronger form.

1.1.1 Definitions

Definition 1.1. *$\emptyset$ is the set having no members.*

This definition bestows the name "$\emptyset$" on a certain set. When we write down such a definition we must be sure of two things: that there exists such a set and that there cannot be more than one set having no members. The empty set axiom provides the first fact and extensionality provides the second. Without both facts the symbol "$\emptyset$" would not be well defined.

Definition 1.2.

- *For any sets u and v, the* ***pair set*** *$\{u, v\}$ is the set whose members are u and v.*
- *For any sets a and b, the* ***union*** *$a \cup b$ is the set whose members are those sets belonging either to a or to b (or both).*
- *For any set a, the* ***power set*** *$\mathscr{P}a$ is the set whose members are exactly the subsets of a.*

The existence and uniqueness proofs guaranteeing all of these definitions can be proved using the empty set axiom, the pairing axiom, the union axiom, and the power set axiom respectively, each in combination with the extensionality axiom.

We can use the pairing and union together to form other finite sets. Given any x we can denote the pair set $\{x, x\}$ as the *singleton* $\{x\}$ (since the pair exists and is unique). For any given x_1, x_2, x_3 we can define

$$\{x_1, x_2, x_3\} = \{x_1, x_2\} \cup \{x_3\}$$

(since the union exists and is unique for any two particular sets). Similarly we can define $\{x_1, x_2, x_3, x_4\}$ and so forth.
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Notation
Our existence axioms have been of the form "there exists a set B whose members are those sets x satisfying the condition __"

$$\exists B \forall x (x \in B \leftrightarrow \_\_)$$

Where if some other sets $t_1, \ldots, t_k$ are involved in the condition, then the full version becomes

$$\forall t_1 \cdots \forall t_k \exists B \forall x (x \in B \leftrightarrow \_\_)$$

With the blank filled in be some expression involving $t_1, \ldots, t_k$ and x. The empty set axiom can be written as

$$\exists B \forall x (x \in B \leftrightarrow x \neq x)$$

Now let us be more general and consider any statement of the form

$$\forall t_1 \cdots \forall t_k \exists B \forall x (x \in B \leftrightarrow \_\_)$$

If the statement were true, then the set B, by the statement itself, as well as by equivalence, can be named by use of the abstraction notation:

$$B = \{x \mid \_\_\}$$

*My interpretation was that showing the existence and uniqueness of B amounts to showing:

$$\forall t_1 \cdots \forall t_k [\exists B (\forall x (x \in B \leftrightarrow \_\_) \wedge \forall B' (\forall x (x \in B' \leftrightarrow \_\_) \rightarrow B' = B))]$$

The sets defined earlier can similarly be named in this notation:

$$\begin{aligned} \emptyset &= \{x \mid x \neq x\} \\ \{u, v\} &= \{x \mid x = u \text{ or } x = v\} \\ a \cup b &= \{x \mid x \in a \text{ or } x \in b\} \\ \mathscr{P} a &= \{x \mid x \subseteq a\} \end{aligned}$$

Note that it is not the case that all statements of the general form are true. For instance

$$\exists B \forall x (x \in B \leftrightarrow x = x)$$

is false as it asserts the existence of a set B to which every set belongs.

1.1.2 Subset axioms

Subset axioms
For each formula __ not containing B, the following is an axiom:

$$\forall t_1 \cdots \forall t_k \forall c \exists B \forall x (x \in B \leftrightarrow x \in c \land \_\_)$$

Given an existing set c, this axiom asserts the existence of a set whose members are exactly those sets in c such that __.

See that such a set must be the subset of an existing set c (thus the name). The set can be named by use of a variation on the abstraction notation:

$$B = \{x \in c \mid \_\_\}$$

Intersection
One of the subset axioms is

$$\forall a \forall c \exists B \forall x (x \in B \leftrightarrow x \in c \land x \in a)$$

The axiom asserts the existence of the set we define to be the *intersection* $c \cap a$ of c and a.

Relative complement
We also have as a subset axiom

$$\forall A \forall B \exists S \forall x (x \in S \leftrightarrow x \in A \land x \notin B)$$

This set S is the *relative complement* of B in A, denoted $A - B$.

1.1.3 Russell's Paradox

Theorem 1.3. *There is no set to which every set belongs.*

Proof. Suppose, for contradiction, that there exists a set A to which every set belongs. By the subset axiom we can assert

$$\forall A \exists B \forall x (x \in B \leftrightarrow x \in A \land x \notin x)$$

That is, the existence of

$$B = \{x \in A \mid x \notin x\}$$

By construction of B we have

$$\forall x (x \in B \leftrightarrow x \in A \land x \notin x)$$

and

$$B \in B \leftrightarrow B \in A \land B \notin B$$

We know that $B \in A$. This can be used to show

$$B \in B \leftrightarrow B \notin B$$ □

The idea here is that if B were not a member of itself, then by definition B should be in B. But if B were a member of itself, then by definition it should not be a member of itself.

1.1.4 Improved Union Axiom

The previously described union operation allowed us to consider the union of two sets, and by repeating the operation, the union of three sets and so on. Suppose however we want the union of infinitely many sets:

$$A = \{b_0, b_1, b_2, \ldots\}$$

For this we need a more general union operation:

$$\begin{aligned}\bigcup A &= \bigcup_i b_i \\ &= \{x \mid x \text{ belongs to some member } b_i \text{ of } A\}\end{aligned}$$

This leads us to define the union of any set A as

$$\begin{aligned}\bigcup A &= \{x \mid x \text{ belongs to some member of } A\} \\ &= \{x \mid (\exists b \in A) x \in b\}\end{aligned}$$

We need an improved union axiom to include cases where A were infinite, overcoming the limitation of defining unions strictly "one by one".

Union axiom
For any set A, there exists a set B whose elements are exactly the members of the members of A:

$$\forall A \exists B \forall x [x \in B \leftrightarrow (\exists b \in A) x \in b]$$

For any A we can define $\bigcup A$ as

$$\bigcup A = \{x \mid (\exists b \in A) x \in b\}$$

See that for any two sets a and b we have

$$\begin{aligned}\bigcup \{a, b\} &= \{x \mid (\exists c \in \{a, b\}) x \in c\} \\ &= \{x \mid x \in a \vee x \in b\} \\ &= a \cup b\end{aligned}$$

(one can check that the $\{x \mid x \in a \vee x \in b\}$ is the equivalent notation for the binary union.) This shows that our preliminary form of the union axiom can be discarded in favor of the new form—the set $a \cup b$ produced by the preliminary form can also be obtained from the pairing and revised form of the union axiom.

1.1.5 Redundance of the preliminary union axiom

Theorem 1.4. *Given the pairing axiom*

$$\forall u \forall v \exists B \forall x (x \in B \leftrightarrow x = u \textit{ or } x = v)$$

and the union axiom

$$\forall A \exists B \forall x [x \in B \leftrightarrow (\exists b \in A) x \in b]$$

we can show

$$\forall a \forall b \exists B \forall x (x \in B \leftrightarrow x \in a \textit{ or } x \in b)$$

which is the preliminary union axiom determined earlier.

Proof. Let a and b be arbitrary sets. The pairing axiom asserts the existence of a set A where

$$\forall x (x \in A \leftrightarrow x = a \text{ or } x = b)$$

Now that we know set A exists, the union axiom axiom asserts the existence of a set B where

$$\forall x [x \in B \leftrightarrow (\exists b \in A) x \in b]$$

Let x be arbitrary.

($\rightarrow$) First suppose $x \in B$. We can then declare some $y \in A$ such that $x \in y$. Since $y \in A$ we have $y = a \vee y = b$, and therefore $x \in a \vee x \in b$.

($\leftarrow$) Now suppose that $x \in a \vee x \in b$. In the case where $x \in a$, we know that $x \in a \in A$, so $x \in B$. The case where $x \in b$ is similarly proven. $\square$

1.1.6 Intersection

Theorem 1.5. *For any nonempty set A, there exists a unique set B such that for any x,*

$$x \in B \leftrightarrow x \textit{ belongs to every member of } A$$

Proof. We first prove existence. We want to show

$$\exists B(\forall x(x \in B \leftrightarrow \forall a \in A(x \in a)))$$

Since A is nonempty we can let c be some member of A. By the subset axiom we can assert

$$\exists B(x \in B \leftrightarrow x \in c \wedge \forall a \in A \backslash \{c\}(x \in a))$$

So we can declare some B such that the above is true. Let x be arbitrary.

($\rightarrow$) First suppose that $x \in B$, then let $a \in A$. We have two cases; first suppose $a = c$, then we know $x \in c = a$. Now suppose that $x \neq c$; then since $x \in A$ we have $x \in A \backslash \{c\}$ and therefore $x \in a$.

($\leftarrow$) Now suppose that $\forall a \in A(x \in a)$. Since $c \in A$, we must have $x \in c$. Let a be some element of $A \backslash \{c\}$; clearly $a \in A$, so $x \in a$. We have now shown $x \in c \wedge \forall a \in A \backslash \{c\}(x \in a)$, so $x \in B$. Existence is proven.

Uniqueness, as always, follows from extensionality. We consider some arbitrary B' and assume that $\forall x(x \in B' \leftrightarrow \forall a \in A(x \in a))$, then show that $B' = B$ using extensionality. □

In contrast to the union operation, no special axiom is needed to justify the intersection operation. This theorem permits defining a unique intersection set for every nonempty set. We proved that for any nonempty A,

$$\exists B(\forall x(x \in B \leftrightarrow \forall a \in A(x \in a)))$$

and that this set was unique for each A. Using the earlier notation we name this set

$$\bigcap A = \{x \mid x \text{ belongs to every member of } A\}$$

An edge case should be considered: our theorem proves existence of the intersection $\bigcap A$ for all nonempty A; what happens in the case where $A = \emptyset$? In other words, is it the case that

$$\exists B(\forall x(x \in B \leftrightarrow \forall a \in \emptyset(x \in a)))$$

We can show that this implies (by vacuous truth)

$$\exists B(\forall x(x \in B))$$

which would contradict russells paradox.

1.2 Set algebra

The union operation for any two sets exists by the pairing, union, and extensionality axioms.

$$A \cup B = \{x \mid x \in A \vee x \in B\}$$

The intersection operation for any two sets exists by the subset and extensionality axioms.

$$A \cap B = \{x \mid x \in A \wedge x \in B\}$$

Also by the subset and extensionality axioms, we have the *relative complement* $A - B$ *of* B *in* A.

$$A - B = \{x \in A \mid x \notin B\}$$

also sometimes denoted as $A \backslash B$.

Note that we cannot form (as a set) the "absolute complement" of B, $\{x \mid x \neq B\}$. In this case the union with B, which must exist axiomatically, would be a set of all sets, which would contradict russell's paradox.

The study of the operations of union ($\cup$), intersection ($\cap$), and relative complementation ($-$), together with the inclusion relation ($\subseteq$), is called the *algebra of sets*. The following identities, which hold for any sets, are some elementary facts of the algebra of sets:

Commutative laws

$$A \cup B = B \cup A \quad \text{and} \quad A \cap B = B \cap A$$

Associative laws

$$A \cup (B \cup C) = (A \cup B) \cup C$$
$$A \cap (B \cap C) = (A \cap B) \cap C$$

Proofs of these follow directly from the commutativity and associativity of the logical OR and AND.
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Distributive laws

$$A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$$
$$A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$$

Proof. First we show $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$. Let x be arbitrary.

($\rightarrow$) Suppose $x \in A \cap (B \cup C)$. Then $x \in A$ and $x \in B \cup C$. In the case where $x \in B$, then $x \in A \cap B$. In the case where $x \in C$, then $x \in A \cap C$. So we have $x \in A \cap B \vee x \in A \cap C$ and $x \in (A \cap B) \cup (A \cap C)$.

($\leftarrow$) Suppose $x \in (A \cap B) \cup (A \cap C)$. Then $x \in (A \cap B) \vee x \in (A \cap C)$. In the case where $x \in (A \cap B)$ then $x \in A$ and $x \in B$. So $x \in B \cup C$ and $x \in A \cap (B \cup C)$. In the case where $x \in (A \cap C)$ then $x \in A$ and $x \in C$. Again $x \in B \cup C$ and $x \in A \cap (B \cup C)$. We are done.

Now we show $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$. Let x be arbitrary.

($\rightarrow$) Suppose $x \in A \cup (B \cap C)$. Then $x \in A$ or $x \in B \cap C$. In the case where $x \in A$ we have both $x \in A \cup B$ and $x \in A \cup C$, so $x \in (A \cup B) \cap (A \cup C)$. In the case where $x \in B \cap C$, then $x \in B$ and $x \in C$. So $x \in (A \cup B)$ and $x \in (A \cup C)$ and $x \in (A \cup B) \cap (A \cup C)$.

($\leftarrow$) Suppose $x \in (A \cup B) \cap (A \cup C)$. We have $x \in (A \cup B)$ and $x \in (A \cup C)$. We have two cases. Suppose first that $x \in A$; then $x \in A \cup (B \cap C)$. Now suppose $x \notin A$. Then since $x \in (A \cup B)$ and $x \in (A \cup C)$ we must have $x \in B$ and $x \in C$. So $x \in B \cap C$ and $x \in A \cup (B \cap C)$. □

De Morgan's laws

$$C - (A \cup B) = (C - A) \cap (C - B)$$
$$C - (A \cap B) = (C - A) \cup (C - B)$$

Proof. First we show $C - (A \cup B) = (C - A) \cap (C - B)$. Let x be arbitrary.

($\rightarrow$) Suppose $x \in C - (A \cup B)$. Then $x \in C$ and $x \notin A \cup B$, meaning $\neg(x \in A \cup B)$, $\neg(x \in A \vee x \in B)$, and $x \notin A \wedge x \notin B$. So since $x \in C$, we have $x \in C - A$ and $x \in C - B$, and therefore $x \in (C - A) \cap (C - B)$.

($\leftarrow$) Suppose $x \in (C - A) \cap (C - B)$. Then $x \in C$, $x \notin A$, and $x \notin B$. We therefore have $\neg(x \in A \vee x \in B)$ and $x \notin A \cup B$. So $x \in C - (A \cup B)$. We are done.

Now we show $C - (A \cap B) = (C - A) \cup (C - B)$. Let x be arbitrary.

($\rightarrow$) Suppose $x \in C - (A \cap B)$. Then $x \in C$ and $\neg(x \in A \wedge x \in B)$. So $x \notin A$ or $x \notin B$. In the case where $x \notin A$, $x \in C - A$. In the case where $x \notin B$, $x \in C - B$. We therefore have $x \in C - A$ or $x \in C - B$, and $x \in (C - A) \cup (C - B)$.

($\leftarrow$) Suppose $x \in (C - A) \cup (C - B)$. We have two cases. First suppose $x \in C - A$, then $x \in C$ and $x \notin A$. The latter implies $\neg(x \in A \wedge x \in B)$ and $x \notin A \cap B$, so $x \in C - (A \cap B)$. Now suppose $x \in C - B$. In a similar manner this implies $x \in C$ and $x \notin A \cap B$, and therefore $x \in C - (A \cap B)$. □

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One generally considers sets as subsets of some larger set or "space" S. Assume then that A and B are subsets of S. We can then abbreviate $S - A$ as simply $-A$, the set S being understood as fixed. In this abbreviation De Morgan's laws become

$$-(A \cup B) = -A \cap -B$$
$$-(A \cap B) = -A \cup -B$$

Identities involving $\emptyset$

$$A \cup \emptyset = A$$
$$A \cap \emptyset = \emptyset$$
$$A \cap (C - A) = \emptyset$$

Proof. We show $A \cup \emptyset = A$. Let x be arbitrary. First suppose that $x \in A \cup \emptyset$. This implies that $x \notin \emptyset \to x \in A$. Therefore $x \in A$. The other direction is clearly true.

Both of the other two statements can be proved by extensionality, where both directions in each statement are vacuously true. $\square$

Inclusion
Under the assumption that $A \subseteq S$ (and that $S - A$ can be abbreviated as $-A$), we have

$$A \cup S = S$$
$$A \cap S = A$$
$$A \cup -A = S$$
$$A \cap -A = \emptyset$$

Proof. First we show $A \cup S = S$. Let x be arbitrary. Suppose that $x \in A \cup S$; then $x \in A$ or $x \in S$. In the case where $x \in A$, since $A \subseteq S$ then $x \in S$. The case where $x \in S$ is self-evident. Now suppose that $x \in S$, then clearly $x \in A \cup S$. We are done.

Next we show $A \cap S = A$. Let x be arbitrary. Suppose $x \in A \cap S$. This clearly implies $x \in A$. Now suppose that $x \in A$. Since $A \subseteq S$ we also have $x \in S$ and therefore $x \in A \cap S$.

To show $A \cup -A = S$, let x be arbitrary and suppose $x \in A \cup -A$. This means $x \in A \vee (x \in S \wedge x \notin A)$. In the case where $x \in A$ we know that $A \subseteq S$ so $x \in S$. The other case where $x \in S \wedge x \notin A$ is self-evident. Now suppose $x \in S$; we have two cases. In the first case suppose $x \in A$. Then clearly $x \in A \cup -A$. In the other case $x \notin A$. Then $x \in S - A$ and $x \in A \cup -A$.

The last statement follows from a previously proven identity. Namely that for any two sets A and C we have $A \cap (C - A) = \emptyset$. $\square$

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Inclusion: Monotonicity properties

$$A \subseteq B \to A \cup C \subseteq B \cup C$$
$$A \subseteq B \to A \cap C \subseteq B \cap C$$
$$A \subseteq B \to \bigcup A \subseteq \bigcup B$$

Proof. For the first property, suppose $A \subseteq B$. Let x be an arbitrary element of $A \cup C$. Then $x \in A$ or $x \in C$. In the case where $x \in A$, since $A \subseteq B$ we also have $x \in B$ and $x \in B \cup C$. In the case where $x \in C$ we have $x \in B \cup C$.

For the second property, again suppose $A \subseteq B$. Let x be an arbitrary element of $A \cap C$. Then $x \in A$ and $x \in C$. Since $A \subseteq B$, $x \in B$; so $x \in B \cap C$.

For the last property, suppose $A \subseteq B$ and let x be an arbitrary element of $\bigcup A$. Then there is some $a \in A$ such that $x \in a$. Since $A \subseteq B$, $x \in a \in B$; so $x \in \bigcup B$. □

Antimonotone properties

$$A \subseteq B \to C - B \subseteq C - A$$
$$\emptyset \neq A \subseteq B \to \bigcap B \subseteq \bigcap A$$

Proof. For the first property, suppose $A \subseteq B$ and let x be an arbitrary element of $C - B$. This means $x \in C$ and $x \notin B$. The contrapositive of the fact that $A \subseteq B$ shows that $x \notin A$. So $x \in C - A$.

For the second property, suppose A is nonempty and $A \subseteq B$ (which also means that B is nonempty). Let x be an arbitrary element of $\bigcap B$. Let a be some element of A. Since $A \subseteq B$, $a \in B$. Since $x \in \bigcap B$ we then have $x \in a$. □

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Distributive laws

$$A \cup \bigcap B = \bigcap\{A \cup X \mid X \in B\} \quad \text{for } B \neq \emptyset$$
$$A \cap \bigcup B = \bigcup\{A \cap X \mid X \in B\}$$

The set $\{A \cup X \mid X \in B\}$ is an extension of the abstraction notation; read "the set of all $A \cup X$ such that $X \in B$". It defines the unique set D such that

$$\forall x(x \in D \leftrightarrow \exists X \in B(x = A \cup X))$$

The existence of such a set D can be proved by observing that $A \cup X \subseteq A \cup \bigcup B$, which means that D is a subset of $\mathscr{P}(A \cup \bigcup B)$. A similar approach can be used to assert the existence of $\{A \cap X \mid X \in B\}$.

Proof. (First statement) First we prove the existence of $\{A \cup X \mid X \in B\}$. We do this by showing that for arbitrary x, A, and B,

$$\exists X \in B(x = A \cup X) \to x \in \mathscr{P}(A \cup \bigcup B)$$

Suppose there exists some $X \in B$ such that $x = A \cup X$. Let y be some arbitrary element of x. Then $y \in A \cup X$. In the case where $y \in A$ we clearly have $y \in A \cup \bigcup B$. In the case where $y \in X$, since $X \in B$ we then have $y \in \bigcup B$ and $y \in A \cup \bigcup B$. This shows that $x \subseteq A \cup \bigcup B$ and $x \in \mathscr{P}(A \cup \bigcup B)$. As such we have

$$\forall A \forall B \forall x[\exists X \in B(x = A \cup X) \leftrightarrow x \in \mathscr{P}(A \cup \bigcup B) \land \exists X \in B(x = A \cup X)]$$

The statement

$$\forall A \forall B \exists C(\forall x(x \in C \leftrightarrow x \in \mathscr{P}(A \cup \bigcup B) \land \exists X \in B(x = A \cup X)))$$

Is true by the subset axiom. These two statements can be used to show

$$\forall A \forall B \exists C(\forall x(x \in C \leftrightarrow \exists X \in B(x = A \cup X)))$$

Proving the existence of $\{A \cup X \mid X \in B\}$ for arbitrary A and B, where uniqueness follows from extensionality.
(next page)

Now we prove the first property.

$$A \cup \bigcap B = \bigcap\{A \cup X \mid X \in B\} \quad \text{for } B \neq \emptyset$$

Let x be arbitrary.

($\rightarrow$) Suppose $x \in A \cup \bigcap B$. This means that either $x \in A$ or $x \in \bigcap B$. Let a be some arbitrary element of $\{A \cup X \mid X \in B\}$; this means that there exists some $X \in B$ such that $a = A \cup X$. In the case where $x \in A$, we clearly have $x \in A \cup X = a$. In the case where $x \in \bigcap B$, since $X \in B$ we must have $x \in X$. Therefore $x \in A \cup X = a$.

($\leftarrow$) Now suppose that $x \in \bigcap\{A \cup X \mid X \in B\}$. This means $\forall a \in \{A \cup X \mid X \in B\}(x \in a)$. We have 2 cases. First suppose $x \in A$; then $x \in A \cup \bigcap B$. Now suppose $x \notin A$; we will show that $x \in \bigcap B$. Let $b \in B$; then we know $A \cup b \in \{A \cup X \mid X \in B\}$. So $x \in A \cup b$. Since $x \notin A$ this must mean that $x \in b$. So $x \in \bigcap B$ and $x \in A \cup \bigcap B$. $\square$

Proof. (Second statement) First we prove the existence of $\{A \cap X \mid x \in B\}$, defined as the unique set C where for arbitrary A and B,

$$\forall x(x \in C \leftrightarrow \exists X \in B(x = A \cap X))$$

We first show

$$\forall A \forall B \forall x(\exists X \in B(x = A \cap X) \rightarrow x \in \mathscr{P}(A \cap \bigcup B))$$

Let A, B, and x be arbitrary. Suppose there exists $X \in B$ such that $x = A \cap X$. Let y be some arbitrary element of x. Then $y \in A$ and $y \in X \in B$, meaning $y \in \bigcup B$. So $y \in A \cup \bigcup B$ and $x \in \mathscr{P}(A \cup \bigcup B)$.

We now know

$$\forall A \forall B \forall x(\exists X \in B(x = A \cap X) \leftrightarrow x \in \mathscr{P}(A \cap \bigcup B) \wedge \exists X \in B(x = A \cap X))$$

By the subset axiom we can assert

$$\forall A \forall B \exists C(\forall x(x \in C \leftrightarrow x \in \mathscr{P}(A \cap \bigcup B) \wedge \exists X \in B(x = A \cap X)))$$

Using these two facts we can prove

$$\forall A \forall B \exists C(\forall x(x \in C \leftrightarrow x \in \exists X \in B(x = A \cap X)))$$

Proving the existence of $\{A \cap X \mid X \in B\}$ for arbitrary A and B, where uniqueness follows from extensionality.
(next page)

Now we prove the second property.

$$A \cap \bigcup B = \bigcup \{A \cap X \mid X \in B\}$$

Let x be arbitrary.

($\rightarrow$) Suppose $x \in A \cap \bigcup B$. Then $x \in A$ and $x \in \bigcup B$, meaning we can declare some $b \in B$ such that $x \in b$. We know $x \in A \cap b$. Since $A \cap b \in \{A \cap X \mid X \in B\}$, we then have $x \in \bigcup \{A \cap X \mid X \in B\}$.

($\leftarrow$) Suppose now that $x \in \bigcup \{A \cap X \mid X \in B\}$ then we can declare some $a \in \{A \cap X \mid X \in B\}$ such that $x \in a$. This means that we can declare some $X \in B$ such that $a = A \cap X$. So $x \in A$ and $x \in X \in B$, which means $x \in \bigcup B$. So $x \in A \cap \bigcup B$. $\square$

De Morgan's laws (for $A \neq \emptyset$)

$$C - \bigcup A = \bigcap \{C - X \mid X \in A\}$$
$$C - \bigcap A = \bigcup \{C - X \mid X \in A\}$$

Existence of $\{C - X \mid X \in A\}$ can be proven by noticing that $X \in A$ implies $\bigcap A \subseteq X$, which implies $C - X \subseteq C - \bigcap A$.

Proof. We first prove existence of $\{C - X \mid X \in A\}$. That is, we prove

$$\forall A \forall C (\exists D \forall x (x \in D \leftrightarrow \exists X \in A (x = C - X)))$$

We start by showing that for nonempty A,

$$\forall A \forall C \forall x (\exists X \in A (x = C - X) \rightarrow x \in \mathscr{P}(C - \bigcap A))$$

Letting A, C, and x be arbitrary, suppose there exists some $X \in A$ such that $x = C - X$. Let y be some arbitrary element of x. Then $y \in C$ and $y \notin X$. For contradiction suppose $y \in \bigcap A$; since $X \in A$ this would mean $y \in X$, which is a contradiction. So $y \in C - \bigcap A$, $x \subseteq C - \bigcap A$, and $x \in \mathscr{P}(C - \bigcap A)$. We are done.

By the subset axiom we can assert

$$\forall A \forall C \exists D \forall x (x \in D \leftrightarrow x \in \mathscr{P}(C - \bigcap A) \wedge \exists X \in A (x = C - X))$$

Using these two facts we can prove

$$\forall A \forall C (\exists D \forall x (x \in D \leftrightarrow \exists X \in A (x = C - X)))$$

Which proves the existence of $\{C - X \mid X \in A\}$ for arbitrary A and C. Uniqness follows from extensionality.
(next page)

Now we prove the first property—that for nonempty A,

$$C - \bigcup A = \bigcap \{C - X \mid X \in A\}$$

Let x be arbitrary.

($\rightarrow$) Suppose $x \in C - \bigcup A$. Then $x \in C$ and $x \notin \bigcup A$. Suppose arbitrary $a \in \{C - X \mid X \in A\}$. Then there exists some $X \in A$ such that $a = C - X$. Suppose for contradiction that $x \in X$; since $X \in A$ this would mean $x \in \bigcup A$, which is a contradiction. Therefore $x \in C - X = a$. Since a was arbitrary we therefore have $x \in \bigcap\{C - X \mid X \in A\}$.

($\leftarrow$) Now suppose $x \in \bigcap\{C - X \mid X \in A\}$. Since A is nonempty we can declare some $X \in A$. Then $C - X \in \{C - X \mid X \in A\}$, and $x \in C - X$. So we have $x \in C$. Now suppose for contradiction that $x \in \bigcup A$. Then we can declare some $X_0 \in A$ such that $x \in X_0$. $C - X_0 \in \{C - X \mid X \in A\}$, so $x \in C - X_0$ and $x \notin X_0$, which is a contradiction. So we have $x \in C - \bigcup A$. This proves the first property.

Now for the second property. For any nonempty A,

$$C - \bigcap A = \bigcup \{C - X \mid X \in A\}$$

Let x be arbitrary

($\rightarrow$) Suppose $x \in C - \bigcap A$. Then $x \in C$ and $x \notin \bigcap A$. The latter implies $\exists a \in A(x \notin a)$. So there is some $a \in A$ such that $x \notin a$. We know $x \in C - a \in \{C - X \mid X \in A\}$; so $x \in \bigcup\{C - X \mid X \in A\}$.

($\leftarrow$) Suppose $x \in \bigcup\{C - X \mid X \in A\}$. Then there exists some $a \in \{C - X \mid X \in A\}$ such that $x \in a$. There then exists some $X \in A$ such that $a = C - X$. So $x \in C$ and $x \notin X$. For contradiction, suppose $x \in \bigcap A$. Since $X \in A$ this would imply that $x \in X$, which is a contradiction. So $x \in C - \bigcap A$. We are done. $\square$

Chapter 2

Relations and Functions

2.1 Ordered Pairs

The pair set $\{a, b\}$ can be thought of as an unordered pair, since $\{a, b\} = \{b, a\}$. We want to define a set $\langle x, y\rangle$ that uniquely encodes both what x, y are, and what order they are in. That is,

$$\forall x \forall y \forall u \forall v (\langle x, y\rangle = \langle u, v\rangle \leftrightarrow (x = u \wedge y = v))$$

Any definition of $\langle x, y\rangle$ that satisfies the above will suffice. For an example of definitions that *fail this requirement*, if we define $\langle x, y\rangle_1 = \{x, y\}$, then for $x \neq y$, $\langle x, y\rangle_1 = \{x, y\} = \{y, x\} = \langle y, x\rangle_1$ even though $x \neq y$.

The most conventional construction was given by Kazimierz Kuratowski in 1921:

Definition 2.1. $\langle x, y\rangle$ *is defined to be* $\{\{x\}, \{x, y\}\}$.

We must prove that this definition succeeds in capturing the desired property.

Theorem 2.2. *For arbitrary* u, v, x, y, $\langle u, v\rangle = \langle x, y\rangle$ *iff* $u = x$ *and* $v = y$.

Proof. The left direction is trivial. If $u = x$ and $v = y$, then $\{\{x\}, \{x, y\}\}$ is identical to $\{\{u\}, \{u, v\}\}$.

For the right direction, suppose $\langle u, v\rangle = \langle x, y\rangle$. We have two cases. First suppose $x = y$. Then $\langle x, y\rangle = \{\{x\}, \{x, y\}\} = \{\{x\}, \{x, x\}\} = \{\{x\}, \{x\}\} = \{\{x\}\}$ and then $\{\{x\}\} = \{\{u\}, \{u, v\}\}$. This means that $\{x\} = \{u, v\}$ and $y = x = u = v$. So clearly $u = x$ and $v = y$. For the other case suppose $x \neq y$. We know $\{\{x\}, \{x, y\}\} = \{\{u\}, \{u, v\}\}$. See that $\{u\} = \{x\} \vee \{u\} = \{x, y\}$; the latter cannot be the case, so we must have $\{u\} = \{x\}$ and $u = x$. We know that $\{x, y\} \neq \{u\}$ since $x \neq y$, so we must have $\{x, y\} = \{u, v\} = \{x, v\}$. This leaves us with $y = x \vee y = v$, so we can conclude that $y = v$. $\square$

This allows us to unambiguously define the *first coordinate* of $\langle x, y\rangle$ to be x and the *second coordinate* to be y.

2.1.1 Cartesian product

Suppose that we have two sets A and B, and we form ordered pairs $\langle x, y\rangle$ with $x \in A$ and $y \in B$. The collection of all such pairs is called the *Cartesian product* $A \times B$ of A and B:

$$A \times B = \{\langle x, y\rangle \mid x \in A \land y \in B\}$$

We must verify that this collection is actually a set before the definition is legal.

Lemma 2.3. *For arbitrary x, y, C, if $x \in C$ and $y \in C$, then $\langle x, y\rangle \in \mathscr{P}\mathscr{P}C$.*

Proof. Suppose $x \in C$ and $y \in C$. Then $\{x\} \subseteq C$ and $\{x, y\} \subseteq C$. So $\{x\} \in \mathscr{P}C$ and $\{x, y\} \in \mathscr{P}C$. So $\{\{x\}, \{x, y\}\} \subseteq \mathscr{P}C$ and $\{\{x\}, \{x, y\}\} \in \mathscr{P}\mathscr{P}C$. □

Now to show existence of $A \times B = \{\langle x, y\rangle \mid x \in A \land y \in B\}$.

Proof. We first show

$$\forall A \forall B \forall t(\exists x \in A \exists y \in B(t = \langle x, y\rangle) \to t \in \mathscr{P}\mathscr{P}(A \cup B))$$

Letting A, B, and t be arbitrary, suppose we can declare $x \in A$ and $y \in B$ such that $t = \langle x, y\rangle$. We have $x \in A \cup B$ and $y \in A \cup B$; by the previous lemma we then have $t = \langle x, y\rangle \in \mathscr{P}\mathscr{P}(A \cup B)$.

By the subset axiom we can assert

$$\forall A \forall B \exists D \forall t(t \in D \leftrightarrow t \in \mathscr{P}\mathscr{P}(A \cup B) \land \exists x \in A \exists y \in B(t = \langle x, y\rangle))$$

Using these two facts we can show

$$\forall A \forall B \exists D \forall t(t \in D \leftrightarrow \exists x \in A \exists y \in B(t = \langle x, y\rangle))$$

Proving the existence of $\{\langle x, y\rangle \mid x \in A \land y \in B\}$ for arbitrary A and B. Uniqueness follows from extensionality. □

2.2 Relations

Definition 2.4. *A **relation** is a set of ordered pairs.*

Any set of ordered pairs is some relation, even if a peculiar one. For a relation R, we sometimes write xRy in place of $\langle x, y\rangle \in R$.

Definition 2.5. *We define the **domain** of R (dom R), the **range** of R (ran R), and the **field** of R (fld R) by*

$$\begin{aligned} x \in dom\, R &\leftrightarrow \exists y(\langle x, y\rangle \in R) \\ x \in ran\, R &\leftrightarrow \exists t(\langle t, x\rangle \in R) \\ fld\, R &= dom\, R \cup ran\, R \end{aligned}$$

Again we must justify the existence of the domain and the range.

Lemma 2.6. *For arbitrary A, x, y, if $\langle x, y\rangle \in A$, then x and y belong to $\bigcup\bigcup A$.*

Proof. We assume $\{\{x\}, \{x, y\}\} \in A$. Consequently, $\{x, y\} \in \bigcup A$ since it belongs to a member of A. In a similar manner we can conclude that $x \in \bigcup\bigcup A$ and $y \in \bigcup\bigcup A$. □

Now to prove existence.

Proof. Starting with the domain, we want to show

$$\forall R \exists D \forall x(x \in D \leftrightarrow \exists y(\langle x, y\rangle \in R))$$

We start by proving

$$\forall R \forall x(\exists y(\langle x, y\rangle \in R) \to x \in \bigcup\bigcup R)$$

Letting R and x be arbitrary, and supposing that there exists y such that $\langle x, y\rangle \in R$, by the previous lemma we have $x \in \bigcup\bigcup R$.

The subset axioms allow us to assert

$$\forall R \exists D \forall x(x \in D \leftrightarrow x \in \bigcup\bigcup R \wedge \exists y(\langle x, y\rangle \in R))$$

Existence follows from these two facts. Uniqueness follows from extensionality.

Now for the range. We want to show

$$\forall R \exists D \forall x(x \in D \leftrightarrow \exists t(\langle t, x\rangle \in R))$$

Again using the previous lemma we can easily prove

$$\forall R \forall x(\exists t(\langle t, x\rangle \in R) \to x \in \bigcup\bigcup R)$$

and then using the subset axiom to assert

$$\forall R \exists D \forall x(x \in D \leftrightarrow x \in \bigcup\bigcup R \wedge \exists t(\langle t, x\rangle \in R))$$

the desired result follows. Uniqueness follows from extensionality. □

2.2.1 n-ary Relations

We can extend the ideas behind ordered pairs to the case of ordered triples and, more generally, to ordered n-tuples. For triples we define

$$\langle x, y, z \rangle = \langle \langle x, y \rangle, z \rangle$$

Similarly we can form ordered quadruples:

$$\langle x_1, x_2, x_3, x_4 \rangle = \langle \langle x_1, x_2, x_3 \rangle, x_4 \rangle = \langle \langle \langle x_1, x_2 \rangle, x_3 \rangle, x_4 \rangle$$

We define an *n-ary relation on A* to be a set of ordered n-tuples with all components in A. Thus a binary (2-ary) relation on A is just a subset of $A \times A$. A ternary (3-ary) relation on A is a subset of $(A \times A) \times A$.

2.3 Functions

Definition 2.7. *A **function** is a relation F such that for each x in $\mathit{dom}\, F$ there is only one y such that xFy.*

$$\forall x \in \mathit{dom}\, F \exists ! y (xFy)$$

*For a function F and a x in $\mathit{dom}\, F$, the unique y such that xFy is called the **value of** F **at** x and is denoted $F(x)$. Thus $\langle x, F(x) \rangle \in F$.*

To avoid confusion, this "$F(x)$" notation should only be used when F is a function and $x \in \operatorname{dom} F$.

Definition 2.8. *For sets A and B, we say that F is a function from A to B or that F maps A into B*

$$F : A \to B$$

iff F is a function, $\mathit{dom}\, F = A$, and $\mathit{ran}\, F \subseteq B$.

Note the unequal treatment of A and B here—we demand only that $\operatorname{ran} F \subseteq B$.

Definition 2.9. *F is a function from A **onto** B if F is a function from A to B, and $\mathit{ran}\, F = B$.*

Thus any function F maps its domain onto its range. The word "onto" depends both on F and on the set B.

Definition 2.10. *A function F is **one-to-one** iff for each $y \in \mathit{ran}\, F$ there is only one x such that xFy.*

$$\forall y \in \mathit{ran}\, F \exists ! x (xFy)$$

*One-to-one functions are sometimes called **injections**.*

(next page)

I use this page to prove a few equivalences of alternative definitions.

Theorem 2.11. *For a function F,*

$$\forall y \in \mathit{ran}\, F \exists! x(xFy) \leftrightarrow \forall x_1 \in \mathit{dom}\, F \forall x_2 \in \mathit{dom}\, F(F(x_1) = F(x_2) \to x_1 = x_2)$$

Proof. ($\to$) Suppose $\forall y \in \operatorname{ran} F \exists! x(xFy)$. This can be rewritten as

$$\forall y \in \operatorname{ran} F(\exists x(xFy) \land \forall x_1 \forall x_2(x_1Fy \land x_2Fy \to x_1 = x_2))$$

Suppose x_1 and x_2 are elements of $\operatorname{dom} F$ such that $F(x_1) = F(x_2)$ (we know that $F(x_1)$ and $F(x_2)$ exist and are unique since F is a function and x_1 and x_2 are in the domain of F.) We know $\langle x_1, F(x_1)\rangle \in F$ and $\langle x_2, F(x_2)\rangle \in F$ so both $F(x_1)$ and $F(x_2)$ are in the range of F. We will denote $F(x_1) = F(x_2) = y$. We then have x_1Fy and x_2Fy for $y \in \operatorname{ran} F$. Using our initial assumption we can conclude that $x_1 = x_2$.

($\leftarrow$) Suppose now that the right side is true. We want to show

$$\forall y \in \operatorname{ran} F(\exists x(xFy) \land \forall x_1 \forall x_2(x_1Fy \land x_2Fy \to x_1 = x_2))$$

Let y be an arbitrary element of $\operatorname{ran} F$. $\exists x(xFy)$ follows by definition. Let x_1 and x_2 be arbitrary such that x_1Fy and x_2Fy. Then x_1 and x_2 are in the domain of F. Since F is a function, $y = F(x_1) = F(x_2)$ it follows from our initial assumption that $x_1 = x_2$. $\square$

Theorem 2.12. *Supposing F is a function from A to B,*

$$\mathit{ran}\, F = B \leftrightarrow \forall y \in B \exists x \in A(xFy)$$

Proof. ($\to$) Suppose $\operatorname{ran} F = B$. Let y be an arbitrary element of B. Then by our assumption, $y \in \operatorname{ran} F$. By definition, there exists some x such that xFy. x is in the domain of F, so since $F : A \to B$, $x \in A$.

($\leftarrow$) Suppose now that $\forall y \in B \exists x \in A(xFy)$. Let y be arbitrary. First suppose $y \in \operatorname{ran} F$. Then since $F : A \to B$, $\operatorname{ran} F \subseteq B$ and $y \in B$. Now suppose $y \in B$. Then by the initial assumption there exists some $x \in A$ such that xFy. This implies that $y \in \operatorname{ran} F$. $\square$

2.3.1 More definitions

It will occasionally be useful to apply the concept of "one-to-one" to relations that are not functions. The phrase seems inapproprate in the context of non-functions, so in such a context we use the phrase "single-rooted".

Definition 2.13. *A set R is single-rooted iff for each $y \in ran\, R$ there is only one x such that xRy.*

$$\forall y \in ran\, F \exists ! x(xRy)$$

For a function, it is single-rooted iff it is one-to-one.

The following can be defined for arbitrary sets A, F, and G:

Definition 2.14.
*The **inverse** of F is the set*

$$F^{-1} = \{\langle u, v\rangle \mid vFu\}$$

*The **composition** of F and G is the set*

$$F \circ G = \{\langle u, v\rangle \mid \exists t(uGt \wedge tFu)\}$$

*The **restriction** of F to A is the set*

$$F \restriction A = \{\langle u, v\rangle \mid uFv \wedge u \in A\}$$

*The **image** of A **under** F is the set*

$$\begin{aligned} F[A] &= \{v \mid \exists u \in A(uFv)\} \\ &= ran(F \restriction A) \end{aligned}$$

In each case we prove existence using the subset axioms.

Proof. (Inverse) We show

$$\forall F \exists D \forall x(x \in D \leftrightarrow \exists u \exists v(x = \langle u, v\rangle \wedge vFu))$$

First we show

$$\forall F \forall x(\exists u \exists v(x = \langle u, v\rangle \wedge vFu) \rightarrow x \in \mathrm{ran}F \times \mathrm{dom}F)$$

Let F and x be arbitrary. Suppose there exists u and v such that $x = \langle u, v\rangle$ and vFu. Then $v \in \mathrm{dom}F$ and $u \in \mathrm{ran}F$, so since $x = \langle u, v\rangle$, $x \in \mathrm{ran}F \times \mathrm{dom}F$.

The subset axiom allows us to assert

$$\forall F \exists D \forall x(x \in D \leftrightarrow x \in \mathrm{ran}F \times \mathrm{dom}F \wedge \exists u \exists v(x = \langle u, v\rangle \wedge vFu))$$

The desired result follows from these two facts. Uniqueness follows from extensionality. ◻

(next page)

Proof. (Composition) We show

$$\forall F \forall G \exists D \forall x (x \in D \leftrightarrow [\exists u \exists v (x = \langle u, v \rangle \wedge \exists t (uGt \wedge tFv))])$$

We first show

$$\forall F \forall G \forall x ([\exists u \exists v (x = \langle u, v \rangle \wedge \exists t (uGt \wedge tFv))] \rightarrow x \in \text{dom} G \times \text{ran} F)$$

Let F, G, and x be arbitrary. Suppose there exists u and v such that $x = \langle u, v \rangle$ and $\exists t (uGt \wedge tFv)$. Then there exists t such that uGt and tFv; this means that $u \in \text{dom} G$ and $v \in \text{ran} F$. So since $x = \langle u, v \rangle$, $x \in \text{dom} G \times \text{ran} F$.

The subset axiom allows us to assert

$$\forall F \forall G \exists D \forall x (x \in D \leftrightarrow x \in \text{dom} G \times \text{ran} F \wedge [\exists u \exists v (x = \langle u, v \rangle \wedge \exists t (uGt \wedge tFv))])$$

The desired result follows from these two facts. Uniqueness follows from extensionality. □

Proof. (Restriction) We show

$$\forall F \forall A \exists D (\forall x (x \in D \leftrightarrow \exists u \exists v (x = \langle u, v \rangle \wedge uFv \wedge u \in A)))$$

It is straightforward to show

$$\forall F \forall A \forall x (\exists u \exists v (x = \langle u, v \rangle \wedge uFv \wedge u \in A) \rightarrow x \in F)$$

Then with the subset axiom we can assert

$$\forall F \forall A \exists D (\forall x (x \in D \leftrightarrow x \in F \wedge \exists u \exists v (x = \langle u, v \rangle \wedge uFv \wedge u \in A)))$$

The desired result follows from these two facts. Uniqueness follows from extensionality. □

Proof. (Image) We show

$$\forall F \forall A \exists D \forall x (x \in D \leftrightarrow \exists u \in A (uFx))$$

It is straightforward to show

$$\forall F \forall A \forall x (\exists u \in A (uFx) \rightarrow x \in \text{ran} F)$$

Then with the subset axiom we can assert

$$\forall F \forall A \exists D \forall x (x \in D \leftrightarrow x \in \text{ran} F \wedge \exists u \in A (uFx))$$

The desired result follows from these two facts. Uniqueness follows from extensionality.
(next page)

Next we show $F[A] = \text{ran}(F \upharpoonright A)$. Let x be arbitrary.

($\rightarrow$) Suppose $x \in F[A]$. This means $\exists u \in A(uFx)$, so there exists some $u \in A$ such that uFx. So $\langle u, x\rangle \in F \upharpoonright A$ and $x \in \text{ran}(F \upharpoonright A)$.

($\leftarrow$) Now suppose $x \in \text{ran}(F \upharpoonright A)$. Then there exists some t such that $\langle t, x\rangle \in F \upharpoonright A$. This means tFx and $t \in A$, so $x \in F[A]$. $\square$

2.3.2 Theorems

Theorem 2.15. *For a set F, $dom\,F^{-1} = ran\,F$ and $ran\,F^{-1} = dom\,F$. For a relation F, $(F^{-1})^{-1} = F$.*

Proof. First to show that for any arbitrary set F, $\text{dom}F^{-1} = \text{ran}F$. Let x be arbitrary and suppose $x \in \text{dom}F^{-1}$. Then there exists some y such that $xF^{-1}y$. By definition this means yFx, so $x \in \text{ran}F$. Now suppose $x \in \text{ran}F$. Then there exists some y such that yFx. By definition this means $xF^{-1}y$, so $x \in \text{dom}F^{-1}$.

Next we show that for any arbitrary set F, $\text{ran}F^{-1} = \text{dom}F$. Let x be arbitrary and suppose $x \in \text{ran}F^{-1}$. Then there exists some y such that $yF^{-1}x$. By definition this means xFy, so $x \in \text{dom}F$. Now suppose $x \in \text{dom}F$. Then there exists some y such that xFy. By definition then $yF^{-1}x$, so $x \in \text{ran}F^{-1}$.

Finally we show that for any arbitrary relation F, $(F^{-1})^{-1} = F$. Let x be arbitrary and suppose $x \in (F^{-1})^{-1}$. Then there exists u and v such that $x = \langle u, v\rangle$ and $vF^{-1}u$. By definition we then have uFv and $x = \langle u, v\rangle \in F$. Now suppose $x \in F$. Since F is a relation x must be an ordered pair, whose first and second coordinates can be denoted as u and v such that $x = \langle u, v\rangle$ and uFv. By definition $vF^{-1}u$ and $u(F^{-1})^{-1}v$, so $x = \langle u, v\rangle \in (F^{-1})^{-1}$. $\square$

Theorem 2.16. *For a set F, F^{-1} is a function iff F is single-rooted. A relation F is a function iff F^{-1} is single-rooted.*

Proof. First we prove that for any set F, F^{-1} is a function iff F is single-rooted.

($\rightarrow$) First suppose F^{-1} is a function. Let y be some element of $\text{ran}F$. By definition there exists some x such that xFy. We also have $yF^{-1}x$ and $y \in \text{dom}F^{-1}$. Now let there be some other x' such that $x'Fy$. Then since xFy and $x'Fy$, $yF^{-1}x$ and $yF^{-1}x'$. Since F^{-1} is a function we then have $x' = x$, so F is single-rooted.

($\leftarrow$) Now suppose F is single-rooted. Let y be some element of $\text{dom}F^{-1}$. By definition this means there exists some x such that $yF^{-1}x$, and xFy. So $y \in \text{ran}F$. Let x_1 and x_2 exist such that $yF^{-1}x_1$ and $yF^{-1}x_2$, meaning x_1Fy and x_2Fy. Since F is single rooted, we must then have $x_1 = x_2$. So F^{-1} is a function.

(next page)

Now we prove that a relation F is a function iff F^{-1} is single-rooted.

($\rightarrow$) Suppose F were a function. Let x be an element of ranF^{-1}. By definition there then exists some y such that $yF^{-1}x$. This also means xFy so $x \in$ dom$F$. Let y_1 and y_2 be arbitrary such that $y_1F^{-1}x$ and $y_2F^{-1}x$. Then xFy_1 and xFy_2. Since F is a function we then have $y_1 = y_2$.

($\leftarrow$) Suppose F^{-1} were single-rooted. Let x be an element of dom$F$. Then by definition there exists some y such that xFy. This means $yF^{-1}x$ so $x \in$ ranF^{-1}. Now let y_1 and y_2 be arbitrary such that xFy_1 and xFy_2. This means $y_1F^{-1}x$ and $y_2F^{-1}x$; since F^{-1} is single-rooted, $y_1 = y_2$. □

Theorem 2.17. *Assume that F is a one-to-one function. If $x \in domF$, then $F^{-1}(F(x)) = x$. If $y \in ranF$, then $F(F^{-1}(y)) = y$.*

Proof. First to show $\forall x(x \in \text{dom}F \rightarrow F^{-1}(F(x)) = x)$. Let x be arbitrary and suppose $x \in$ dom$F$. Then there exists some y such that $\langle x, y\rangle \in F$; by definition this y is unique and $y = F(x)$. By definition $\langle y, x\rangle \in F^{-1}$. Since F is one-to-one F^{-1} is a function and $x = F^{-1}(y) = F^{-1}(F(x))$.

Now to show $\forall y(y \in \text{ran}F \rightarrow F(F^{-1}(y)) = y)$. Let y be arbitrary and suppose $y \in$ ran$F$. Then by definition there exists a unique x such that $\langle x, y\rangle \in F$, and $y = F(x)$. We also have $\langle y, x\rangle \in F^{-1}$; since F^{-1} is a function $x = F^{-1}(y)$. So $y = F(x) = F(F^{-1}(y))$. □

Theorem 2.18. *Assume F and G are functions. Then $F \circ G$ is a function. Its domain is*

$$\{x \in domG \mid G(x) \in domF\}$$

and for x in its domain $(F \circ G)(x) = F(G(x))$.

Proof. Assuming F and G are functions. First we show that $F \circ G$ is a function. Suppose $x \in \text{dom}(F \circ G)$. Then by definition there exists y such that $\langle x, y\rangle \in F \circ G$. Suppose y_1 and y_2 are arbitrary elements such that $\langle x, y_1\rangle \in F \circ G$ and $\langle x, y_2\rangle \in F \circ G$; then by definition there exists t_1 and t_2 such that $\langle x, t_1\rangle \in G$, $\langle t_1, y_1\rangle \in F$, $\langle x, t_2\rangle \in G$, and $\langle t_2, y_2\rangle \in F$. Since G is a function we then have $t_1 = t_2$. So $\langle t_1, y_1\rangle \in F$ and $\langle t_1, y_2\rangle \in F$. Since F is a function we then have $y_1 = y_2$.

Now we show $\text{dom}(F \circ G) = \{x \in \text{dom}G \mid G(x) \in \text{dom}F\}$. We can assert the existence of the right side by the subset axiom:

$$\exists D(\forall x(x \in D \leftrightarrow x \in \text{dom}G \wedge G(x) \in \text{dom}F))$$

$G(x)$ exists since G is a function and dom$G$ and dom$F$ are sets.

($\rightarrow$) Suppose $x \in \text{dom}(F \circ G)$. Then by definition there exists y such that $\langle x, y\rangle \in F \circ G$; then there exists t such that $\langle x, t\rangle \in G$ and $\langle t, y\rangle \in F$. Then $x \in$ dom$G$ and since G is a function $G(x) = t \in$ dom$F$.

($\leftarrow$) Suppose now $x \in \{x \in \text{dom}G \mid G(x) \in \text{dom}F\}$. Then $x \in$ dom$G$ and $G(x) \in$ dom$F$. The former means there exists t such that xGt, so $t = G(x)$. This, combined with the latter means $t \in$ dom$F$, so there exists y such that tFy. By definition we then have $\langle x, y\rangle \in F \circ G$ and $x \in \text{dom}(F \circ G)$.
(next page)

Finally we show that $\forall x \in \mathrm{dom}(F \circ G)[(F \circ G)(x) = F(G(x))]$. Let x be an arbitrary element of $\mathrm{dom}(F \circ G)$. Then there exists y such that $\langle x, y\rangle \in F \circ G$ Since $F \circ G$ is a function $y = (F \circ G)(x)$. By definition there exists t such that xGt and tFy. Since G is a function $t = G(x)$ and $y = F(t)$. Putting it all together we have $(F \circ G)(x) = y = F(t) = F(G(x))$. □

2.3.3 Identity relation

Definition 2.19. *For a set ω, the identity function on ω is*

$$I_\omega = \{\langle n, n\rangle \mid n \in \omega\}$$

Proof. (Existence) We want to show

$$\forall \omega \exists D(\forall x(x \in D \leftrightarrow \exists n \in \omega(x = \langle n, n\rangle)))$$

First we show

$$\forall \omega \forall x(\exists n \in \omega(x = \langle n, n\rangle) \to x \in \omega \times \omega)$$

Let ω and x be arbitrary. Suppose there exists some $n \in \omega$ such that $x = \langle n, n\rangle$. Then by definition $x \in \omega \times \omega$. By the subset axiom we can assert

$$\forall \omega \exists D(\forall x(x \in D \leftrightarrow x \in \omega \times \omega \wedge \exists n \in \omega(x = \langle n, n\rangle)))$$

where the desired conclusion follows from these two facts. Uniqueness follows from extensionality. □

Proof. (I_ω is a function on ω) We first show that I_ω is a function. Let $x \in \mathrm{dom}I_\omega$. Then by definition there exists y such that $\langle x, y\rangle \in I_\omega$. Let y_1 and y_2 be arbitrary such that $\langle x, y_1\rangle \in I_\omega$ and $\langle x, y_2\rangle \in I_\omega$. Then there exists n_1 and n_2 such that $\langle x, y_1\rangle = \langle n_1, n_1\rangle$ and $\langle x, y_2\rangle = \langle n_2, n_2\rangle$. So $n_1 = x = n_2$ and $y_1 = n_1 = n_2 = y_2$.

Next we show $\mathrm{dom}I_\omega = \omega$. Let $x \in \mathrm{dom}I_\omega$. Then there exists some y such that $\langle x, y\rangle \in I_\omega$. By definition there exists some $n \in \omega$ such that $\langle x, y\rangle = \langle n, n\rangle$. So $x = n \in \omega$. Now let $x \in \omega$. By the pairing axiom $\langle x, x\rangle$ exists and by definition is an element of I_ω. So $x \in \mathrm{dom}I_\omega$.

Finally we show $\mathrm{ran}I_\omega = \omega$. Let $x \in \mathrm{ran}I_\omega$. Then there exists y such that $\langle y, x\rangle \in I_\omega$. By definition there then exists $n \in \omega$ such that $\langle n, n\rangle = \langle y, x\rangle$, and $x = n \in \omega$. Now let $x \in \omega$; then $\langle x, x\rangle$ exists and by definition is an element of I_ω. So $x \in \mathrm{ran}I_\omega$. □

2.3.4 More theorems, Axiom of Choice first form

Theorem 2.20. *Assuming that G is some one-to-one function, $G^{-1} \circ G$ is the identity function on dom$G$, I_{domG}, and $G \circ G^{-1}$ is the identity on ran$G$, I_{ranG}.*

Proof. Assuming that G is some one-to-one function, we first show $G^{-1} \circ G$ is the identity function on dom$G$, $I_{\text{dom}G}$. Since G is a one-to-one function, by theorem 2.16 G^{-1} is also a one-to-one function; by theorem 2.18 $G^{-1} \circ G$ is a function.

We can show that $\text{dom}(G^{-1} \circ G) = \text{dom}G$. By theorem 2.18, the domain of $G^{-1} \circ G$ is given by

$$\{x \in \text{dom}G \mid G(x) \in \text{dom}G^{-1}\}$$

Suppose $x \in \text{dom}(G^{-1} \circ G)$. Then by the above equivalence it immediately follows that $x \in \text{dom}G$. Now suppose $x \in \text{dom}G$. Then there exists some y such that xGy and $yG^{-1}x$. This means $G(x) = y \in \text{dom}G^{-1}$. By the above equivalence $x \in \{x \in \text{dom}G \mid G(x) \in \text{dom}G^{-1}\} = \text{dom}(G^{-1} \circ G)$.

Now to show $G^{-1} \circ G = I_{\text{dom}G}$. Let $\langle x, y\rangle$ be an arbitrary element of $G^{-1} \circ G$. Then $x \in \text{dom}(G^{-1} \circ G) = \text{dom}G$. By theorem 2.18 $(G^{-1} \circ G)(x) = G^{-1}(G(x))$. By theorem 2.17 $G^{-1}(G(x)) = x$. Putting it all together we have $y = (G^{-1} \circ G)(x) = G^{-1}(G(x)) = x$, so $\langle x, y\rangle = \langle x, x\rangle$ for $x \in \text{dom}G$, so by definition $\langle x, y\rangle \in I_{\text{dom}G}$.

Now suppose $\langle x, y\rangle \in I_{\text{dom}G}$. Then there exists some $n \in \text{dom}G$ such that $\langle x, y\rangle = \langle n, n\rangle$. This implies $x = n = y$ and $x \in \text{dom}G = \text{dom}(G^{-1} \circ G)$ in a manner similar to the previous direction $(G^{-1} \circ G)(x) = G^{-1}(G(x)) = x = y$. So $\langle x, y\rangle \in G^{-1} \circ G$.

Next we show $G \circ G^{-1}$ is the identity on ran$G$, $I_{\text{ran}G}$. First we show that $\text{dom}(G \circ G^{-1}) = \text{ran}G$. By theorem 2.18 $G \circ G^{-1}$ is a function and the domain of $G \circ G^{-1}$ is equivalent to

$$\{x \in \text{dom}G^{-1} \mid G^{-1}(x) \in \text{dom}G\}$$

Suppose $x \in \text{dom}(G \circ G^{-1})$. Then by the above equivalence and theorem 2.15 we immediately have $x \in \text{dom}G^{-1} = \text{ran}G$. Now suppose $x \in \text{ran}G$. Then $x \in \text{dom}G^{-1}$ and there exists some y such that yGx and $xG^{-1}y$. So $G^{-1}(x) = y \in \text{dom}G$. By the above equivalence we then have $x \in \text{dom}(G \circ G^{-1})$.

Suppose $\langle x, y\rangle \in G \circ G^{-1}$. Then $x \in \text{dom}(G \circ G^{-1}) = \text{ran}G$. By theorem 2.18 we have $y = (G \circ G^{-1})(x) = G(G^{-1}(x))$. Since G is a one-to-one function and $x \in \text{ran}G$, by theorem 2.17 $G(G^{-1}(x)) = x$. Putting it all together we have $y = (G \circ G^{-1})(x) = G(G^{-1}(x)) = x$. So $\langle x, y\rangle = \langle x, x\rangle$ for $x \in \text{ran}G$. By definition we have $\langle x, y\rangle \in I_{\text{ran}G}$.

Now suppose $\langle x, y\rangle \in I_{\text{ran}G}$. By definition there exists some $n \in \text{ran}G$ such that $\langle x, y\rangle = \langle n, n\rangle$. This implies $x = n = y$ and $x \in \text{ran}G = \text{dom}(G \circ G^{-1})$. Then in a manner similar to the previous direction, $(G \circ G^{-1})(x) = G(G^{-1}(x)) = x = y$. So $\langle x, y\rangle \in G \circ G^{-1}$. □

(next page)

Theorem 2.21. *For any sets F and G,*

$$(F \circ G)^{-1} = G^{-1} \circ F^{-1}$$

Proof. Both $(F \circ G)^{-1}$ and $G^{-1} \circ F^{-1}$ are, by definition, relations. We have the string of equivalences for arbitrary $\langle x, y\rangle$.

$$\begin{aligned}
\langle x, y\rangle \in (F \circ G)^{-1} &\leftrightarrow \langle y, x\rangle \in F \circ G \\
&\leftrightarrow \exists t(yGt \wedge tFx) \\
&\leftrightarrow \exists t(xF^{-1}t \wedge tG^{-1}y) \\
&\leftrightarrow \langle x, y\rangle \in G^{-1} \circ F^{-1}
\end{aligned}$$

$\square$

Theorem 2.22. *Assume that $F : A \to B$, and that A is nonempty.*

1. *There exists a function $G : B \to A$ (a "left inverse") such that $G \circ F$ is the identity function I_A on A iff F is one-to-one.*

2. *There exists a function $H : B \to A$ (a "right inverse") such that $F \circ H$ is the identity function I_B on B iff F is maps A onto B.*

Proof. (1.)

($\to$) Suppose there exists some $G : B \to A$ such that $G \circ F$ is the identity function I_A on A. Let y be an element of ran$F$. Then by definition there exists x such that xFy. Now let x' be arbitrary such that $x'Fy$. We know that $y \in \text{ran}F \subseteq B = \text{dom}G$. By theorem 2.18 $x \in \text{dom}(G \circ F)$ and $G(y) = G(F(x)) = (G \circ F)(x)$ since $G \circ F$ is the identity function I_A we then have $(G \circ F)(x) = x$. A similar argument shows $G(y) = G(F(x')) = (G \circ F)(x') = x'$. So yGx' and yGx; since G is a function $x' = x$.

($\leftarrow$) Suppose now that F is one-to-one. We need to come up with some function G that satisfies the following requirements: G is a function, dom$G = B$, ran$G \subseteq A$, and $G \circ F = I_A$. Since F is one-to-one F^{-1} is a function from ran$F$ onto A. By assumption A is nonempty, so we can fix some a in A, defining G such that it assigns a to every point in $B - \text{ran}F$ (so that the domain of G encompasses all of B):

$$G(x) = \begin{cases} F^{-1}(x) & \text{if } x \in \text{ran}F \\ a & \text{if } x \in B - \text{ran}F \end{cases}$$

More precisely, our candidate function is

$$G = F^{-1} \cup (B - \text{ran}F) \times \{a\}$$

(next page)

We show that G, defined as

$$G = F^{-1} \cup (B - \text{ran}F) \times \{a\}$$

is, firstly, a function. Let x be an arbitrary element of dom$G$. By definition then exists y such that xGy. Now let y' be arbitrary such that xGy'. We have four cases.

Case 1: $\langle x, y\rangle \in F^{-1}$, $\langle x, y'\rangle \in F^{-1}$. Then since F^{-1} is a function, $y = y'$.

Case 2: $\langle x, y\rangle \in F^{-1}$, $\langle x, y'\rangle \in (B - \text{ran}F) \times \{a\}$. In this case $x \in (B - \text{ran}F)$ so $x \notin \text{ran}F$. But since $\langle x, y\rangle \in F^{-1}$ we also have $\langle y, x\rangle \in F$ and $x \in \text{ran}F$. This case leads to a contradiction and cannot occur; it is excluded.

Case 3: $\langle x, y\rangle \in (B - \text{ran}F) \times \{a\}$, $\langle x, y'\rangle \in F^{-1}$. In a manner similar to case 2, this case leads to a contradiction and can be excluded.

Case 4: $\langle x, y\rangle \in (B - \text{ran}F) \times \{a\}$, $\langle x, y'\rangle \in (B - \text{ran}F) \times \{a\}$. Then $y = a = y'$.

Next we show dom$G = B$. First suppose $x \in \text{dom}G$. Then there exists y such that $\langle x, y\rangle \in G$, meaning $\langle x, y\rangle \in F^{-1}$ or $\langle x, y\rangle \in (B - \text{ran}F) \times \{a\}$. In the first case $\langle y, x\rangle \in F$ so $x \in \text{ran}F \subseteq B$ and $x \in B$. In the second case it immediately follows that $x \in B$.

Now suppose $x \in B$. We have two cases. First suppose $x \in \text{ran}F$. Then there exists y such that $xF^{-1}y$; so xGy and $x \in \text{dom}G$. Next suppose $x \notin \text{ran}F$. Then it immediately follows that $x \in B - \text{ran}F$; $\langle x, a\rangle$ exists by the pairing axiom and is a member of $(B - \text{ran}F) \times \{a\}$, so $\langle x, a\rangle \in G$ and $x \in \text{dom}G$.

Next we show ran$G \subseteq A$. Suppose $x \in \text{ran}G$. Then there exists y such that $\langle y, x\rangle \in F^{-1}$ or $\langle y, x\rangle \in (B - \text{ran}F) \times \{a\}$. In the first case $\langle x, y\rangle \in F$ and $x \in \text{dom}F$. In the second case it immediately follows that $x = a \in A$.

Finally we show $G \circ F = I_A$. Both $G \circ F$ and I_A are relations. Suppose $\langle x, y\rangle$ is an arbitrary element of $G \circ F$. Then there exists t such that xFt and tGy. The latter implies $\langle t, y\rangle \in F^{-1}$ or $\langle t, y\rangle \in (B - \text{ran}F) \times \{a\}$. In the first case yFt and xFt; so since F is one-to-one $x = y$. Since $\langle x, y\rangle = \langle x, x\rangle$ for $x \in \text{dom}F = A$, by definition $\langle x, y\rangle \in I_A$. In the other case where $\langle t, y\rangle \in (B - \text{ran}F) \times \{a\}$ we have $t \notin \text{ran}F$. But since xFt we also have $t \in \text{ran}F$. This case leads to a contradiction and can be excluded.

Suppose now that $\langle x, y\rangle$ is an arbitrary element of I_A. Then there exists some $z \in A$ such that $\langle x, y\rangle = \langle z, z\rangle$. Since $A = \text{dom}F$, there exists z_0 such that zFz_0; it follows that $z_0 \in \text{ran}F \subseteq B$ and $z_0 \in B = \text{dom}G$. Then since G was proven to be a function there exists z_1 such that z_0Gz_1. We have $\langle z_0, z_1\rangle \in F^{-1}$ or $\langle z_0, z_1\rangle \in (B - \text{ran}F) \times \{a\}$ since $z_0 \in \text{ran}F$ the latter case is false and $\langle z_0, z_1\rangle \in F^{-1}$. We then have zFz_0 and z_1Fz_0; since F is one-to-one, $z_1 = z$. So since zFz_0 and $z_0Gz_1 \leftrightarrow z_0Gz$, $\langle x, y\rangle = \langle z, z\rangle \in G \circ F$. $\square$

(next page)

Intuition

The left direction of the second proof poses a difficulty. Assuming F maps A onto B, F not necessarily being one-to-one means we cannot guarantee F^{-1} is a function. So we cannot take $H = F^{-1}$ leaving us with limited options for a candidate H. The idea is that for each $y \in B$ we must *choose* some x for which $F(x) = y$ and let $H(y)$ be the chosen x. Since F is onto we know that every y has one or more x such that $F(x) = y$; we just have no way of defining any one particular choice of x. The axiom of choice gives us a way out.

Axiom of Choice (first form)
For any relation R there is a function $H \subseteq R$ with $\mathrm{dom}H = \mathrm{dom}R$.

$$\forall R \exists H (H \subseteq R \land H \text{ is a function} \land \mathrm{dom}H = \mathrm{dom}R)$$

This axiom asserts that one can "choose" exactly one output or every input in a relation's domain.

Proof. (2)

($\rightarrow$) Suppose there exists some $H : B \rightarrow A$ such that $F \circ H = I_B$. First suppose that $x \in \mathrm{ran}F$; it immediately follows from $\mathrm{ran}F \subseteq B$ that $x \in B$. Next suppose $x \in B$. Then $x \in \mathrm{dom}H$ so there exists some y such that xHy. $y \in \mathrm{ran}H \subseteq A = \mathrm{dom}F$ then implies the existence of z such that yFz. We then have $\langle x, z\rangle \in F \circ H = I_B$, implying the existence of $n \in B$ such that $\langle x, z\rangle = \langle n, n\rangle$, where it then follows that $x = n = z$. So $yFz \leftrightarrow yFx$ and $x \in \mathrm{ran}F$.

($\leftarrow$) Suppose F maps A onto B, meaning $\mathrm{ran}F = B$. F^{-1} is a relation. By the axiom of choice we can assert the existence of a function H such that $H \subseteq F^{-1}$ and $\mathrm{dom}H = \mathrm{dom}F^{-1} = \mathrm{ran}F = B$.

Next we show $\mathrm{ran}H \subseteq A$. Let $y \in \mathrm{ran}H$. Then there exists x such that xHy. Then $xF^{-1}y$, $y \in \mathrm{ran}F^{-1} = \mathrm{dom}F = A$.

Finally we show $F \circ H = I_B$. Let $\langle x, y\rangle$ be an arbitrary element of $F \circ H$. Then there exists t such that xHt and tFy. The former implies $xF^{-1}t$ and tFx. This, combined with the latter and the fact that F is a function, means $x = y$. We have $\langle x, y\rangle = \langle x, x\rangle$ for $x \in \mathrm{ran}F = B$, so $\langle x, y\rangle \in I_B$.

Now suppose $\langle x, y\rangle \in I_B$. Then there exists $n \in B$ such that $\langle x, y\rangle = \langle n, n\rangle$, and it follows that $x = n = y$. We showed that H is a function from B into A. Since $n \in B = \mathrm{dom}H$ there exists n_0 such that nHn_0. Then $n_0 \in \mathrm{ran}H \subseteq A = \mathrm{dom}F$, so there exists n_1 such that n_0Fn_1. nHn_0 implies $nF^{-1}n_0$ and n_0Fn, so since F is a function $n = n_1$. nHn_0 and n_0Fn_1 means $\langle n, n_1\rangle \in F \circ H$. So $\langle x, y\rangle = \langle n, n\rangle = \langle n, n_1\rangle \in F \circ H$. $\square$

2.3.5 Even more theorems

Theorem 2.23. *The following hold for any sets. (F need not be a function)*

1. *The image of a union is the union of the images:*

$$F[A \cup B] = F[A] \cup F[B] \quad \textit{and} \quad F[\bigcup X] = \bigcup \{F[A] \mid A \in X\}$$

2. *The image of an intersection is included in the intersection of the images:*

$$F[A \cap B] \subseteq F[A] \cap F[B] \quad \textit{and} \quad F[\bigcap X] \subseteq \bigcap \{F[A] \mid A \in X\}$$

   for nonempty X. Equality holds if F is single-rooted.

3. *The image of a difference includes the difference of the images:*

$$F[A] - F[B] \subseteq F[A - B]$$

   Equality holds if F is single-rooted.

Proof. (1)
To prove $F[A \cup B] = F[A] \cup F[B]$ for arbitrary F, A and B we have the following string of equivalences for arbitrary y:

$$\begin{aligned} y \in F[A \cup B] &\leftrightarrow \exists u \in A \cup B(uFy) \\ &\leftrightarrow \exists u \in A(uFy) \vee \exists u \in B(uFy) \\ &\leftrightarrow y \in F[A] \vee y \in F[B] \\ &\leftrightarrow y \in F[A] \cup F[B] \end{aligned}$$

Proving $F[\bigcup X] = \bigcup\{F[A] \mid A \in X\}$ for arbitrary F and X follows a similar principle. Suppose $y \in F[\bigcup X]$. Then by definition there exists some $u \in \bigcup X$ such that uFy. There then also exists $A \in X$ such that $u \in A$, so by definition $y \in F[A]$. Since $A \in X$, $F[A] \in \{F[A] \mid A \in X\}$, so $y \in \bigcup\{F[A] \mid A \in X\}$.

Now suppose $y \in \bigcup\{F[A] \mid A \in X\}$. Then there exists some $B \in \{F[A] \mid A \in X\}$ such that $y \in B$. So there exists $A \in X$ such that $B = F[A]$, and $y \in F[A]$. This means that there exists $u \in A$ such that uFy. Since $A \in X$ we then have $u \in \bigcup X$ and $y \in F[\bigcup X]$. □

(next page)

Proof. (2)

To prove $F[A \cap B] \subseteq F[A] \cap F[B]$ for arbitrary F, A and B, suppose $y \in F[A \cap B]$. Then there exists some $u \in A \cap B$ such that uFy. Since $u \in A$ and $u \in B$ then $y \in F[A]$ and $y \in F[B]$. So $y \in F[A] \cap F[B]$.

The converse, $F[A] \cap F[B] \subseteq F[A \cap B]$, can be proven if F were single rooted. Suppose $y \in (F[A] \cap F[B])$. Then $y \in F[A]$ and $y \in F[B]$, and there exists $u_1 \in A$ and $u_2 \in B$ such that u_1Fy and u_2Fy. Since F is single rooted, we then have $u_1 = u_2$ and $u_1 \in A \cap B$. So $y \in F[A \cap B]$.

Now for the general case. To prove $F[\bigcap X] \subseteq \bigcap\{F[A] \mid A \in X\}$ for arbitrary F and nonempty X, suppose $y \in F[\bigcap X]$. Then there exists $u \in \bigcap X$ such that uFy. Let B be an arbitrary element of $\{F[A] \mid A \in X\}$, then there exists some $A \in X$ such that $B = F[A]$. Since $A \in X$ then $u \in A$ and $y \in F[A] = B$. So $y \in \bigcap\{F[A] \mid A \in X\}$.

The converse, $\bigcap\{F[A] \mid A \in X\} \subseteq F[\bigcap X]$, is again true if F were single rooted. Suppose $y \in \bigcap\{F[A] \mid A \in X\}$. Since X is nonempty then there exists $A \in X$. We know by the subset axiom that $F[A]$ exists, so $F[A] \in \{F[A] \mid A \in X\}$ and $y \in F[A]$. Then there exists $u \in A$ such that uFy. Let B be an arbitrary element of X. Then $F[B] \in \{F[A] \mid A \in X\}$ and $y \in F[B]$. So there then exists $u_1 \in B$ such that u_1Fy. Since F is single rooted we must then have $u = u_1$ and $u \in B$; so $u \in \bigcap X$. Since uFy we then have $y \in F[\bigcap X]$. $\square$

Proof. (3)

To prove, for arbitrary F, A and B that $F[A] - F[B] \subseteq F[A - B]$, suppose $y \in F[A] - F[B]$. Then $y \in F[A]$ and $y \notin F[B]$. There exists some $u \in A$ such that uFy. Suppose, for contradiction, that $u \in B$. Then uFy and $y \in F[B]$. So we must have $u \in A - B$ and $y \in F[A - B]$.

The converse is true if F is single-rooted. Suppose $y \in F[A-B]$. Then there exists $u \in A - B$ such that uFy. $u \in A$ so $y \in F[A]$. For contradiction suppose $y \in F[B]$. Then there exists $v \in B$ such that vFy. Since F is single-rooted we would then have $u = v$ and $u \in B$. So $y \notin F[B]$ and $y \in F[A] - F[B]$. $\square$